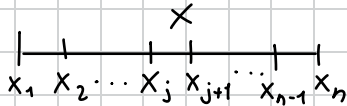


Nyttig for i løse oppgaver:

Vis at med x_i $i=0, \dots, n$ ekvidisterte noder

$$\text{d2} \quad \prod_{i=0}^n |x - x_i| \leq \frac{1}{4} h^{n+1} \cdot n! \quad h = x_i - x_{i-1}$$

Bevis



$$\text{Anta } x_j \leq x \leq x_{j+1}$$

$$\text{d2} \quad |x - x_j| |x - x_{j+1}| \leq \frac{h^2}{4}$$

$$|x - x_j| |x - x_{j+1}| = (x - x_j)(x_{j+1} - x) \quad \text{max verdi: f2s i } \frac{x + x_{j+1}}{2}$$

$$\prod_{i=0}^n |x - x_i| \leq \frac{h^2}{4} \cdot \prod_{i=0}^{j-1} (x - x_i) \prod_{i=j+2}^n (x_i - x) \leq \frac{h^2}{4} \prod_{i=0}^{j-1} (x_{j+1} - x_i) \prod_{i=j+2}^n (x_i - x_j)$$

$$x_i = z + ih$$

$$x_{j+1} - x_i = (j - i + 1)h$$

$$x_i - x_j = (i - j)h$$

$$\leq \underbrace{\frac{h^2}{4} h^j h^{n-(j+2)+1}}_{\frac{h^{n+1}}{4}} \prod_{i=0}^{j-1} (j - i + 1) \prod_{i=j+2}^n (i - j) = \frac{h^{n+1}}{4} (j+1)! (n-j)! \leq \frac{h^{n+1}}{4} n!$$

$$\frac{n!}{k!(n-k)!} \geq 1$$

$$n! \geq k!(n-k)!$$

Forelesning

NUMERISK INTEGRASJON

Numeriske metoder for å beregne integraler (kap 4.4)

Vi skal approksimere $\int_a^b f(x) dx$

Bruk en enkel ide: 1. Approksimer f med interpolasjonspolynom $f \approx p_n(x)$
på $n+1$ noder
2. Integrer polynomet $\int_a^b f(x) dx \approx \int_a^b p(x) dx$

Anta et $x_i = a + hi$ $i = 0, \dots, n$

$$h = \frac{b-a}{n}$$

$$p_n(x) = \sum_{k=0}^n u_k(x) f(x_k)$$

$$u_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{(x-x_j)}{(x_k-x_j)}$$

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k) \text{ der } w_k = \int_a^b u_k(x) dx$$

Newton-Cotes formel

$x_0, x_1, \dots, x_n$ noder av kvadratur formel
 $w_0, w_1, \dots, w_n$ vekt

Eksempler

1) Trapesmetode

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Her bruker man interpolasjon med lineære polynomer

Vi ser på $p_1(x)$ som interpolerer

$$\begin{array}{c|c} a & b \\ \hline f(a) & f(b) \end{array}$$

$$p_1(x) = \frac{f(b) - f(a)}{b-a} (x-a) + f(a)$$

$$\int_a^b p_1(x) dx = (b-a)f(a) + \frac{f(b) - f(a)}{(b-a)} \int_a^b (x-a) dx$$

$$= (b-a)f(a) + \frac{f(b) - f(a)}{(b-a)} \frac{(b-a)^2}{2}$$

$$= \frac{(b-a)}{2} (f(a) + f(b))$$

2) Simpson-metoden

$$f \approx p_2(x) \quad \text{Bruk} \quad \begin{array}{c|c|c} x_0 & x_1 & x_2 \\ \hline 2 & \frac{2+b}{2} & b \\ \hline y & f(2) & f(\frac{2+b}{2}) & f(b) \end{array}$$

$$y_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$$

$$w_i = \int_2^b y_i(x) dx$$

$$y_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$$

$$w_0 = \frac{b-2}{6} \quad w_1 = \frac{2}{3}(b-2) \quad w_2 = \frac{b-2}{6}$$

$$y_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$\int_2^b f(x) dx \approx \frac{b-2}{6} [f(2) + 4f(\frac{2+b}{2}) + f(b)]$$

Feil formel for Kvadratur formel som kommer fra interpolasjonspolynom

$$E_n(f) := \int_2^b f(x) dx - \sum_{k=0}^n w_k f(x_k) \quad \leftarrow \int_2^b \sum_{k=0}^n y_k(x) f(x_k) dx$$

$$E_n(f) = \int_2^b f(x) - \sum_{k=0}^n (y_k(x) f(x_k)) dx = \int_2^b \frac{f^{(n+1)}(\xi(x))}{(n+1)!} w_{n+1}(x) dx$$

$$\text{Ant2 } f \in C^{n+1}([2, b])$$

$$w_{n+1}(x) = \prod_{j=0}^n (x-x_j)$$

$$|E_n(f)| \leq \frac{M_{n+1}}{(n+1)!} \int_2^b |w_{n+1}(x)| dx$$

$$M_{n+1} = \max_{\xi \in (2, b)} |f^{(n+1)}(\xi)|$$

Eks: Trapes-metode

$$|E_1(f)| \leq \frac{M_2}{2} \int_2^b |x-2||x-b| dx = \frac{M_2}{2} \frac{1}{6} (b-2)^3$$

$$M_2 = \max_{\xi \in (2, b)} |f''(\xi)| = \frac{M_2}{12} (b-2)^3$$

Eks: Simpson

$$|E_2(f)| \leq \frac{M_3}{6} \int_2^b |(x-2)(x-\frac{2+b}{2})(x-b)| dx$$

$$\leq \frac{M_3}{192} (b-2)^4 \quad \text{overestimerer feilen}$$

Bedre feilskranke

$$f \in C^4([2, b])$$

$$E_2(f) = -\frac{(b-2)^5}{2880} f^{(4)}(\xi) \quad \xi \in (2, b)$$

Sammensett kvadratur (Ch 4.3)

Strategi: vi deler opp $[a, b]$ i m subintervaller også bruker vi Newton-Cotes formel på hver subintervall

Sammensett trapes $h = \frac{b-a}{m}$ $x_i = a + ih = a + i \frac{b-a}{m}$

$$\int_a^b f(x) dx = \sum_{i=1}^m \int_{x_{i-1}}^{x_i} f(x_i) dx \quad \text{bruk trapes på } [x_{i-1}, x_i]$$

$$\int_{x_{i-1}}^{x_i} f(x) dx \approx \frac{h}{2} [f(x_{i-1}) + f(x_i)] \quad \int_a^b f(x) dx \approx \frac{h}{2} \sum_{i=1}^m [f(x_{i-1}) + f(x_i)]$$

$$\frac{h}{2} \sum_{i=1}^m [f(x_{i-1}) + f(x_i)] = h \left[\frac{f(x_0)}{2} + f(x_1) + \dots + f(x_{m-1}) + \frac{f(x_m)}{2} \right]$$

$$E_1(f) = \int_a^b f(x) dx - h \left[\frac{f(x_0)}{2} + f(x_1) + \dots + f(x_{m-1}) + \frac{f(x_m)}{2} \right]$$

$$|E_1(f)| \leq \frac{mh^3}{12} M_2 = \frac{(b-a)^3}{m^2 12} M_2$$

$$M_2 = \max_{\xi \in [a, b]} |f''(\xi)|$$

Sammensett Simpson

$[a, b]$ delt i m subintervaller

$$h = \frac{b-a}{2m} \quad x_i = a + ih \quad i = 0, \dots, 2m$$

$$\sum_{i=1}^m \int_{x_{2i-2}}^{x_{2i}} f(x) dx \approx \sum_{i=1}^m \frac{2h}{6} [f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})]$$

$$= \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 2f(x_{2m-2}) + 4f(x_{2m-1}) + f(x_{2m})]$$

første og siste element
ingen vekt

$$|E_2(f)| \leq \frac{M_4}{2880} \frac{(b-a)^5}{m^4}$$

Forelesning

NUMERISK INTEGRASJON

Metoder basert på interpolasjon (interpolatoriske)

$$\int_a^b f(x) dx \quad \text{først } f(x) \approx p_n(x) \text{ interp. polynom}$$

p2 $x_0, x_1, \dots, x_n$ i $[a, b]$
s2 $\int_a^b p_n(x) dx \approx \int_a^b f(x) dx$

$$\sum_{i=0}^n w_i f(x_i) \quad w_i = \int_a^b u_i(x) dx$$

Eksempel

1) Trapezmetode

p_n grad 1

Feilestimet

$$|E_1(f)| < \frac{M_2}{12} (b-a)^3$$

2) Simpsons metode

p_n grad 2

Feilutvikling

$$E_2(f) = \frac{1}{90} \left(\frac{b-a}{2} \right)^5 f^{(4)}(\xi) \quad \xi \in (a, b)$$

Sammensatt trapes

m subintervaller av $[a, b]$ $x_i = a + ih$ $h = \frac{b-a}{m}$

$$E_1(f) = \int_a^b f(x) dx - h \left[\frac{f(x_0) + f(x_1) + \dots + f(x_{n-1}) + f(x_n)}{2} \right]$$

$$|E_1(f)| \leq \frac{M_2}{12} m \left(\frac{b-a}{m} \right)^3 = \frac{M_2}{12} \frac{(b-a)^2}{m^2}$$

M_2 er tredje derivert av f

Sammensatt Simpson

m subintervaller $h = \frac{b-a}{2m}$ $x_i = a + ih$

$$E_2(f) = \int_a^b f(x) dx - \frac{h}{3} \left[f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{2n-1}) + f(x_{2n}) \right]$$

$$\sum_{i=1}^m \frac{h}{3} [f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})]$$

$$|E_2(f)| \leq M_4 \frac{1}{90} \left(\frac{b-a}{2m} \right)^5 m$$

$$= M_4 \frac{1}{2880} \frac{(b-a)^5}{m^4}$$

Midtpunkt formel

1. på $[a, b]$

$$\int_a^b f(x) dx \approx (b-a) f\left(\frac{a+b}{2}\right)$$

2. Sammensatt midtpunkt

$$\int_a^b f(x) dx \approx h \sum_{i=1}^m f(\bar{x}_i) \quad \begin{array}{l} m \text{ subintervaller} \\ \bar{x}_i = \frac{x_{i-1} + x_i}{2} \end{array} \quad h = \frac{b-a}{m}$$

Kan vise at

$$E_1(f) = \frac{(b-a)^2}{24} f''(\xi) \quad \xi \in (a, b)$$

Definisjon - Grad av nøyaktighet

Er største positive heltall $r \geq 2$ slik at kvadratur formel er eksakt for polynomer av grad $\leq r$ (dvs for $p \in \mathbb{P}_r$)

- Trapez $r=1$
- Midtpunkt $r=1$
- Simpson $r=3$

Trazesmetode

$$\int_a^b f(x) dx \approx \frac{b-a}{2} (f(a) + f(b))$$

Trenger å sjekke at kvadraturformel er eksakt på en basis av $\mathbb{P}_1$ dvs. $\{1, x\}$ og ikke får en basis av $\mathbb{P}_2$.

- Konstanter $\int_a^b 1 dx = b-a$ $\frac{b-a}{2} (1+1) = b-a$
 $f(x)=1$

- $f(x)=x$ $\int_a^b x dx = \frac{b^2-a^2}{2}$ $\frac{b-a}{2} (a+b) = \frac{b^2-a^2}{2}$

Siste skritt: Sjekke at det ikke er høyere grad enn 1

- $f(x)=x^2$ $\int_a^b x^2 dx = \frac{b^3-a^3}{3} \neq \frac{b-a}{2} (a^2+b^2)$

Simpsons

Har grad av nøyaktighet 3

Vi skal sjekke $\{1, x, x^2, x^3\}$ og i tillegg x^4

for å se at
↓ det ikke stemmer

- $\int_a^b 1 dx = b-a$ $\frac{b-a}{6} (1+4+1) = b-a$

- $\int_a^b x^2 dx = \frac{b^3-a^3}{3}$ $\frac{b-a}{6} \left[a^2 + 4\left(\frac{a+b}{2}\right)^2 + b^2 \right] = \frac{b^3-a^3}{3}$

- $\int_a^b x dx = \frac{b^2-a^2}{2}$ $\frac{b-a}{6} \left[a + 4\frac{a+b}{2} + b \right] = \frac{b^2-a^2}{2}$

- $\int_a^b x^3 dx = \frac{b^4-a^4}{4}$ $\frac{b-a}{6} \left[a^3 + 4\left(\frac{a+b}{2}\right)^3 + b^3 \right] = \frac{b^4-a^4}{4}$

$$\cdot \int_2^b x^4 dx = \frac{b^5 - 2^5}{5} \neq \frac{b-2}{6} \left[2^4 + 4 \left(\frac{2+b}{2} \right)^4 + b^4 \right]$$

Stemmer ikke med kvadraturformel

Adaptiv simpsons kvadratur

$$I = \int_2^b f(x) dx$$

$$S(2,b) = \frac{b-2}{6} \left[f(2) + 4f\left(\frac{2+b}{2}\right) + f(b) \right]$$

$$E(2,b) = \frac{1}{90} \left(\frac{b-2}{2} \right)^3 f^{(4)}(\xi) \quad \xi \in (2,b)$$

$$I = S(2,b) + E(2,b)$$

Mål med adaptiv kvadratur

Gitt en toleranse TOL finn $\tilde{I} \approx I$ s.t. $|I - \tilde{I}| \leq TOL$ (M2)

For å oppnå dette

1. Oppdel $[2,b]$ i subintervaller (adaptiv)
2. Bruk Simpson på hver intervall
3. Stop når $|I - \tilde{I}| \leq TOL$

VIKTIG

Vi må finne en måte å estimere feilen underveis

MERKNAD

$$Ant2 [2,b] = [2_1, b_1] \cup [2_2, b_2] \dots \cup [2_n, b_n]$$

disjunkt union

$$I|_{[2_i, b_i]} = \int_{2_i}^{b_i} f(x) dx$$

$$|I|_{[2_i, b_i]} - S(2_i, b_i)| \leq TOL \frac{b_i - 2_i}{b-2}$$

(Må krevne dette for hvert subintervall)

$$\Rightarrow \left| I - \sum_{i=1}^n S(2_i, b_i) \right| \leq TOL \underbrace{\sum_{i=1}^n \frac{b_i - 2_i}{b-2}}_{=1} = TOL \frac{b-2}{b-2} = TOL$$

Regning

$$I_0 = S(2,b) \quad E_0 = E(2,b)$$

$$I = I_0 + E_0$$

midtpunkt

$$c = \frac{2+b}{2}$$

✓ kvadratur

$$\tilde{I} = S(2,c) + S(c,b)$$

$$\tilde{E} = E(2,c) + E(c,b)$$

$$I = \tilde{I} + \tilde{E} \quad h = b - 2$$

$$\tilde{E} = -\frac{1}{90} \left(\frac{h}{4}\right)^5 \underbrace{2(f^{(4)}(\xi_1) + f^{(4)}(\xi_2))}_2 \quad \begin{matrix} \xi_1 \in (2,c) \\ \xi_2 \in (c,b) \end{matrix}$$

Middelverdisetning

$$\exists \xi \in (2,b) \text{ s.t. } \underbrace{f^{(4)}(\xi_1) + f^{(4)}(\xi_2)}_2 = f^{(4)}(\xi)$$

Sammenlign

$$E(2,b) = \frac{1}{90} \left(\frac{b-2}{2}\right)^5 f^{(4)}(\xi) \quad \xi \in (2,b)$$

$$\tilde{E} = \frac{1}{90} \left(\frac{b-2}{4}\right)^5 2 f^{(4)}(\xi) = \frac{1}{90} \frac{1}{16} \left(\frac{b-2}{2}\right)^5 f^{(4)}(\xi)$$

Antz $f^{(4)}$ er "nesten" konstant

$$f^{(4)}(\xi) \approx f^{(4)}(\xi) \quad \text{d.t. } E_0 \approx 16\tilde{E}$$

$$\begin{aligned} \text{Da } I &= I_0 + E_0 \Rightarrow \tilde{I} - I_0 = E_0 - \tilde{E} \\ I &= \tilde{I} + \tilde{E} \end{aligned}$$

$$\begin{aligned} \tilde{I} - I_0 &= E_0 - \tilde{E} \\ &\approx 16\tilde{E} - \tilde{E} \\ &\approx 15\tilde{E} \end{aligned}$$

$$|\tilde{E}| \approx \frac{|\tilde{I} - I_0|}{15} \quad \text{praktisk feilestimat}$$

noe jeg kan beregne med egen maskin

Algoritme for adaptiv Simpson (rekursiv algoritme)

adaptiveS(f, 2, b, TOL)

Jupyter notebook

$$I_0 = S(f, 2, b)$$

$$c = \frac{2+b}{2}$$

$$\tilde{I} = S(f, 2, c) + S(f, c, b)$$

$$\text{if } \frac{|\tilde{I} - I_0|}{15} \leq \text{TOL}$$

$$\tilde{I} = \tilde{I} + \frac{\tilde{I} - I_0}{15}$$

else

$$\tilde{I} = \text{adaptiveS}(f, 2, c, \text{TOL}/2) + \text{adaptiveS}(f, c, b, \text{TOL}/2)$$

end

return $\tilde{I}$